A Variational Trust-Region Framework for Learning-Rate Annealing

Oluwasegun Somefun *somefuno.oregonstate.edu*
V John Mathews *mathews@oregonstate.edu*
School of Electrical Engineering and Computer Science,
Oregon State University.

Abstract

This paper develops a variational trust-region framework for deriving learning-rate schedules for stochastic gradient learning. Each iteration of the stochastic gradient algorithm is formulated as a trust-region subproblem that bounds the parameter increment at each iteration. Under a fixed trust-region radius, this yields an iteration-dependent learning rate inversely proportional to the square-root of the second moment of the gradient, recovering the core structure of adaptive moment estimation. When the trust-region radius is allowed to vary and vanish over a finite learning window, learning-rate design becomes a variational optimization problem posed directly on the evolution of the trust-region radius. We show that linear decay and cosine annealing arise as unique minimizers of Dirichlet and regularized Dirichlet energy functionals of the trust-region radius subject to fixed boundary values. The resulting formulation generalizes naturally to multi-point boundary-value problems, yielding warmup-decay, warmup-stable-decay, and related schedules as solutions of a common first-order variational principle. We further characterize smoothness and stability properties associated with vanishing trust-region windows. Together, these results provide a unified foundation for learning-rate annealing and its coupling with moment estimation in stochastic gradient methods, showing that popular learning-rate schedules can be interpreted as boundary-constrained trust-region trajectories whose form is determined by a first-order variational energy principle.

1 Introduction

Learning-rate schedules play a central role in the practical performance of stochastic gradient methods (Umeda & Iiduka, 2024; Bergsma et al., 2024), especially in large-scale deep learning where training can exceed millions of iterations (Caron et al., 2021; Bello et al., 2021; Goujaud et al., 2022). Despite their widespread use, commonly adopted annealing rules like linear decay and cosine schedules (He et al., 2019; Hägele et al., 2024) are typically motivated heuristically and empirically (Ge et al., 2018; Agarwal et al., 2021; Bergsma et al., 2024) rather than derived from first principles. At the same time, popular stochastic gradient methods like RMSProp (Tieleman & Hinton, 2012; Graves et al., 2013) or Adam (Kingma & Ba, 2015) rely on gradient moment-based normalization and learning-rate annealing (Loshchilov & Hutter, 2017; 2019) but lack a unifying theoretical interpretation.

A substantial empirical literature already documents the influence of learning-rate annealing on large-scale training performance (Bergsma et al., 2024; Hägele et al., 2024). In contrast, this paper develops a trust-region framework that unifies second-moment normalization and annealing schedules in stochastic gradient methods. Here, we interpret each iteration of the stochastic gradient method as a trust-region minimization subproblem on the parameter update step. Under a fixed trust-region radius, this formulation yields an optimal iteration-dependent learning rate that is inversely proportional to the square-root of the second moment of the gradient, revealing the underlying structure of adaptive moment estimation in accelerated learning. We then extend this result to a more general case where the trust-region radius of the learning algorithm is varying and vanishes across a finite training window of iterations $t \in [0, \tau]$, and pose variational problems over the trust-region radius to yield its smoothest possible variation under well-defined boundary conditions. This leads to two optimal annealing schedules: linear decay, cosine annealing, together with their root forms, each arising as the extremal solution to a total variational problem governing the smooth contraction of the trust region.

These results provide principled derivations of some of the most widely used annealing schedules in deep learning (Huang et al., 2017; Loshchilov & Hutter, 2017; 2019; Bergsma et al., 2024), showing they are not arbitrary heuristics but arise as optimal solutions to well-posed smoothness-constrained optimization problems.

The main contributions of this paper are as follows:

(1) We develop a trust-region formulation of stochastic gradient learning in which second-moment estimation arises as a learning-rate mechanism from a fixed trust-region radius, while learning-rate annealing arises from a varying and vanishing trust-region radius.

(2) We show that popular learning-rate schedules are unique solutions of first-order variational trust-region subproblems, obtained by minimizing Dirichlet and regularized Dirichlet energies under fixed boundary values.

(3) We characterize the general boundary-value form of the resulting total variational problem, showing that warmup-decay, warmup-stable-decay, and related schedules are generated by boundary value conditions, then we establish general smoothness and stability properties of the resulting solutions with respect to the Dirichlet energy over the trust-region window.

Beyond recovering common learning-rate schedules, these characterizations reveal their functional role within stochastic gradient learning. A constant trust-region radius yields bounded update magnitudes, whereas the derived learning-rate schedules induce exponential mean-square contraction of the parameter update-step vector. Therefore, learning-rate annealing arises from first principles not merely as a step-size reduction strategy, but as a variational trust-region mechanism for converting bounded update-step behavior into an exponentially vanishing update-step vector, thereby enforcing asymptotic stability of the update-step sequence over the learning window.

The remainder of the paper develops these results in detail. Section 2 formulates the vanishing trust-region problem and derives the associated optimal learning-rate function. Sections 2.1–2.2 derive linear decay, square-root linear decay, and cosine annealing as solutions of variational trust-region problems over a finite learning window. Section 2.4 studies structural properties of the resulting window functions, while Section 3 extends the analysis to multi-point boundary-value problems, yielding warmup-decay and warmup-stable-decay schedule families. Further, Sections 3.4 to 3.5 characterize the smoothness of window functions through a normalized Dirichlet-energy factor and establish exponential mean-square stability properties associated with the vanishing trust-region window. Finally, this paper is concluded in Section 4.

2 A Trust-region Problem

In this section, we begin by formulating a trust-region problem (Parikh & Boyd, 2014) on the parameter iterates of the learning algorithm. Let the shorthand notation (t, i) denote the i-th element of an n-dimensional vector at iteration t. Expressed coordinate-wise, the stochastic gradient update algorithm is

$$\mathrm{w}(t+1, i) = \mathrm{w}(t, i) - \alpha(t, i)\, \mathrm{g}(t, i) \tag{1}$$

where $\mathrm{w}(t, i)$ is the i-th coordinate of the parameter vector at iteration t, $\mathrm{g}(t, i)$ is the corresponding stochastic gradient, and $\alpha(t, i) > 0$ is the possibly iteration-dependent learning rate applied to that coordinate.

Throughout, we use $\mathbb{E}[\cdot]$ to denote expectation with respect to the underlying stochastic process generating the gradients. No specific objective function, probabilistic structure, or distribution of the gradients is assumed. We make only the following minimal assumptions for all $t \in \{0, \ldots, \tau\}$ within a finite window of learning iterations where $\tau \gg 0$:

Assumption 1 (Lipschitz Regularity)**.** *The scalar-valued objective function $f\big(\mathrm{w}(t, i)\big)$ and its gradient are Lipschitz continuous (Bottou et al., 2018).*

Assumption 2 (Bounded Second Moment)**.** *For each coordinate i, the stochastic gradient admits a bounded second moment*

$$\mathbb{E}\big[\mathrm{g}^2(t,i)\big] \leq \bar{\xi} < \infty, \quad \textit{where} \quad \bar{\xi} := \sup_{0\leq t\leq \tau} \mathbb{E}[\mathrm{g}^2(t,i)]$$

is a finite and strictly positive constant.

Under this setup for the learning algorithm, we define the parameter's step-change as $\Delta(t+1,i) = \mathrm{w}(t+1,i) - \mathrm{w}(t,i)$. Then from (1), for each coordinate i, the expected squared step-change (second moment),

$$\mathbb{E}[\Delta^2(t+1,i)] = \alpha^2(t,i)\,\mathbb{E}[\mathrm{g}^2(t,i)] \tag{2}$$

is well-defined for all t. This variable is used as a trust-region measure in this analysis. For notational convenience, we define the trust-region measure $\mathbb{E}[\Delta^2(t+1,i)]$ of the algorithm as an iterative function $\delta^2(t)$, that is $\delta^2(t) \equiv \mathbb{E}[\Delta^2(t+1,i)]$. Therefore (2) is equivalent to

$$\alpha^2(t,i)\,\mathbb{E}[\mathrm{g}^2(t,i)] = \delta^2(t), \tag{3}$$

where $\delta(t)$ is the iterative trust-region radius of the learning algorithm. At each learning iteration of (1), it is desired that its step-changes are bounded within a maximum trust region radius $\rho \in \big(0,1\big)$, such that

$$\delta^2(t) = \mathbb{E}[\Delta^2(t+1,i)] \;\leq\; \rho^2, \tag{4}$$

and that $\delta(t) \to 0$ as $t \to \tau$.

Such behavior is desirable for several reasons. One reason is that the asymptotic stability of the parameter step, $\Delta(t+1,i) \to 0$ becomes artificially guaranteed both in the mean and in the mean-square sense. Consequently, the parameter iterates do not oscillate indefinitely due to gradient noise and as $t \to \tau$, the update step becomes increasingly conservative near a local optimum. This is a classical stability requirement in stochastic approximation (Ge et al., 2018). We will refer to this as the *vanishing trust-region radius* problem.

From (3), it can be observed that the varying trust-region radius, $\delta(t)$ directly determines the learning-rate, that is

$$\alpha(t,i) = \frac{\delta(t)}{\sqrt{\mathbb{E}[\mathrm{g}^2(t,i)]}}. \tag{5}$$

In the absence of the vanishing trust-region constraint, when the function $\delta(t)$ is fixed at an initial value, specifically $\delta(t) = \mu$ for some positive scalar $\mu \leq \rho$, the trust-region radius remains constant. In this case, the learning rate recovers the core structure of adaptive moment estimation, and the overall trust-region constraint,

$$\mathbb{E}\big[\|\Delta(t+1)\|_2^2\big] = n\,\mu^2 \tag{6}$$

is satisfied. This learning-rate can also be derived as the solution to a first-order trust-region subproblem, analogous to the Cauchy point in classical trust-region methods (Nocedal & Wright, 2006),

$$\min_{\alpha(t,i)} \; \mathbb{E}[\Delta(t+1,i)\,\mathrm{g}(t,i)] \quad \text{s.t.} \quad \mathbb{E}[\Delta^2(t+1,i)] = \delta^2(t). \tag{7}$$

Under a first-order Taylor's series approximation of $f(\mathrm{w}(t))$ along coordinate i, $f(\mathrm{w}(t+1,i)) - f(\mathrm{w}(t,i)) \approx \mathrm{g}(t,i)\,\Delta(t+1,i)$. Taking expectations yields $\mathbb{E}[f(\mathrm{w}(t+1,i)) - f(\mathrm{w}(t,i))] \;\approx\; \mathbb{E}[\mathrm{g}(t,i)\,\Delta(t+1,i)]$. Therefore, (7) is a local surrogate of the expected variation of the objective function, along a coordinate i.

In contrast to a fixed trust-region radius of the parameter update step $\delta(t) = \mu$, the vanishing trust-region radius corresponds to fixed boundary conditions $\delta(0) = \mu$, $\delta(\tau) = 0$. To obtain a unique closed-form solution of $\delta(t)$ that produces the smoothest possible contraction of the trust-region radius, we now pose total variational minimization problems (also referred to as Dirichlet energy minimization (Iwaniec & Onninen, 2022)) directly on the vanishing trust-region radius $\delta(t)$ (or its squared radius, $\delta^2(t)$). We will see that each problem enforces a different notion of smoothness that forces the variable trust-region radius not to vanish abruptly or faster than necessary during learning.

2.1 Linear Decay via Dirichlet Energy Minimization of $\delta(t)$

Under fixed boundary conditions $\delta(0) = \mu$, $\delta(\tau) = 0$, we now find the $\delta(t)$ that minimize an energy function on the total trust-region variation between iterations over the window of iterations $t \in \{0, \dots, \tau\}$. That is, we solve for $\delta(t)$ via the optimization problem

$$\min_{0 \le \delta(t) \le \mu} \mathcal{D} \triangleq \sum_{t=1}^{\tau} \big(\delta(t) - \delta(t-1)\big)^2. \tag{8}$$

In the minimization problem (8), $\delta(t) - \delta(t-1)$ is the first-order difference of $\delta(t)$ and measures its rate of change over iterations t, therefore minimizing the total sum of squared differences in (8) forces $\delta(t)$ to be as close as possible to $\delta(t-1)$ across iterations.

The formulation in (8) is strictly convex in $\delta(t)$, and its minimizer is unique. For any interior index $k \in \{1, \dots, \tau-1\}$, taking the derivative $d\,\mathcal{D}_1/\delta(k)$ and setting it to 0 results in $2\big(\delta(k)-\delta(k-1)\big)-2\big(\delta(k+1)-\delta(k)\big) = 0$ which is equivalent to the second-order homogenous linear recurrence or difference equation

$$\delta(k+1) = 2\,\delta(k) - \delta(k-1) \tag{9}$$

For the recurrence in (9), its characteristic equation is $z^2 - 2\,z + 1 = (z-1)^2 = 0$. It has two real and equal roots $z_{1,2} = 1$. Therefore, $\forall\, t$, the general solution to (9) must be of the form

$$\delta(t) = a + b\,t, \tag{10}$$

where $a, b \in \mathbb{R}$. Plugging in the fixed boundary conditions $\delta(0) = \mu$, $\delta(\tau) = 0$, at $t = 0$, and $t = \tau$, into (10) yields

$$\begin{aligned} \mu &= a \\ 0 &= a + \tau\,b, \end{aligned} \tag{11}$$

which leads to both $b = -\mu/\tau$, and $a = \mu$. The complete solution to (9) is then

$$\delta(t) = \mu\left(1 - \frac{t}{\tau}\right), \tag{12}$$

and plugging (12) in (8), the minimum energy in the total variation of $\delta(t)$ is $\mu^2\,\tau^{-1}$. The unique solution to (8) in (12) is the *linear decay* function. In addition to the vanishing trust-region, the trust-region constraint is always satisfied since from (12), $\delta^2(t) \le \mu^2$, implying $\rho = \mu$.

2.2 Square-root Linear Decay via Dirichlet Energy Minimization of $\delta^2(t)$

In the first formulation (8), the total trust-region variational problems between iterations over the window $t \in \{0, \dots, \tau\}$ was posed on $\delta(t)$. The same energy minimization problem may be posed directly on $\delta^2(t) \equiv \mathbb{E}[\Delta^2(t+1, i)]$, under the fixed boundary conditions $\delta^2(0) = \mu^2$, $\delta^2(\tau) = 0$, such that we now find $\delta(t)$ through the $\delta^2(t)$ that minimize

$$\min_{0 \le \delta^2(t) \le \mu^2} \sum_{t=1}^{\tau} \big(\delta^2(t) - \delta^2(t-1)\big)^2, \tag{13}$$

where $\delta^2(t) - \delta^2(t-1)$ is the first-order difference of $\delta^2(t)$, Since (13) is strictly convex in $\delta^2(t)$, so its minimizer is unique. Applying the same steps used in solving (8), we obtain the complete solution to (13) as

$$\delta^2(t) = \mu^2\left(1 - \frac{t}{\tau}\right) \tag{14}$$

to obtain

$$\delta(t) = \mu\sqrt{1 - \frac{t}{\tau}}. \tag{15}$$

The unique solution to (13) in (15) is the square-root of the *linear decay* function obtained in (12). Plugging (15) in (13), the minimum energy in the total variation of $\delta^2(t)$ is $\mu^4\,\tau^{-1}$, and similarly, the trust-region constraint is always satisfied since from (14), $\delta^2(t) \le \mu^2$, and implies $\rho = \mu$.

2.3 Cosine Annealing via a Regularized Dirichlet Energy Minimization of $\delta(t)$

In the previous sections, we saw that minimizing an energy function (8) on the total trust-region variation between iterations over the window $t \in \{0, \dots, \tau\}$ under fixed boundary conditions led to the linear decay function. For convenience, from (10) we can express the general solution of $\delta(t)$ in affine form

$$\delta(t) = a + b\, u(t) \tag{16}$$

where $u(t)$ is an iterative function of t, and $a, b \in \mathbb{R}$, and the fixed boundary conditions $\delta(0) = \mu$, $\delta(\tau) = 0$ correspond to $u(0) = \frac{\mu - a}{b}$, and $u(\tau) = -\frac{a}{b}$. This shows that the non-trivial underlying functional form of $\delta(t)$ is captured by a shaping function $u(t)$ while a, b are trivial shift and scaling constants determined by the fixed boundary conditions. The minimization problem of $\delta(t)$, then becomes equivalent to

$$\min_{u(t) \,\in\, [u(0),\, u(\tau)]} \quad \sum_{t=1}^{\tau} \big(u(t) - u(t-1)\big)^2. \tag{17}$$

Here, we now define a regularized minimization problem directly on $u(t)$. Due to fixed boundary conditions, the variational objective term $\big(u(t) - u(t-1)\big)^2$ is essentially defined over transition pairs $(t-1,\, t)$ within the interior of the window $t \in \{1, \dots, \tau - 1\}$. Therefore, any added quadratic energy regularization should be expressed in a compatible pairwise form in order to maintain consistency with the transition-based structure of the objective term (Grady & Polimeni, 2010).

A natural choice is the symmetric quadratic form $\frac{1}{2}\big(u^2(t) + u^2(t-1)\big)$, which is analogous to Tikhonov regularization and introduces a shape-smoothing penalty (Yue et al., 2016) into (17) in a manner consistent with its pairwise structure. We therefore define the regularized energy minimization problem as

$$\min_{u(t) \,\in\, [u(0),\, u(\tau)]} \quad \mathcal{B} \triangleq \sum_{t=1}^{\tau} \big(u(t) - u(t-1)\big)^2 + \lambda \left[\sum_{t=1}^{\tau} \tfrac{1}{2}\big(u^2(t) + u^2(t-1)\big) - \tilde{c} \right], \tag{18}$$

where $\lambda \in \mathbb{R}$ is the penalty constant or Lagrange multiplier associated with the regularization term, and $\tilde{c} > 0$ is a fixed constant representing the allowable total quadratic energy.

Using the affine form of the general solution (16) makes (18) a regularized minimization problem of $\delta(t)$. Solving for $\delta(t)$ via (18) using (16) simplifies the derivation of regularized variants of (8) without altering the problem in terms of $\delta(t)$. Substituting $u(t) = (\delta(t) - a)/b$ into (18) yields

$$\min_{0 \le \delta(t) \le \mu} \quad \sum_{t=1}^{\tau} \frac{1}{b^2}\big(\delta(t) - \delta(t-1)\big)^2 + \lambda \left[\sum_{t=1}^{\tau} \frac{1}{2b^2}\big(\delta^2(t) + \delta^2(t-1)\big) + \frac{1}{b^2} \sum_{t=1}^{\tau} \big(\delta(t) + \delta(t-1)\big) + \frac{a^2}{b^2} - \tilde{c} \right]. \tag{19}$$

Therefore, since (18) is strongly convex, its minimizer is also unique. For the initial boundary condition, $t = 0$, taking the derivative $d\,\mathcal{B}/u(0) = 0$ gives $-2\big(u(1) - u(0)\big) + \lambda\, u(0) = 0$, which is equivalent to

$$u(1) - u(0) = \tfrac{\lambda}{2}\, u(0). \tag{20}$$

For the final boundary condition, $t = \tau$, taking the derivative $d\,\mathcal{B}/u(\tau) = 0$ gives $2\big(u(\tau) - u(\tau - 1)\big) + \lambda\, u(\tau) = 0$ which is equivalent to

$$u(\tau) - u(\tau - 1) = -\tfrac{\lambda}{2}\, u(\tau). \tag{21}$$

For any interior index $k \in \{1, \dots, \tau - 1\}$, taking the derivative $d\,\mathcal{B}/u(k) = 0$ gives $2\big(u(k) - u(k-1)\big) - 2\big(u(k+1) - u(k)\big) + 2\lambda\, u(k) = 0$, which simplifies to

$$u(k+1) = (2 + \lambda)\, u(k) - u(k-1). \tag{22}$$

For the difference equation in (22), its characteristic equation is $z^2 - (2 + \lambda)\, z + 1 = 0$. The roots of this equation are $z_{1,2} = \frac{(2+\lambda) \pm \sqrt{(2+\lambda)^2 - 4}}{2}$, implying that the characteristic equation can be written as $z^2 - (z_1 + z_2)\, z + (z_1 z_2) = 0$. From the solution for z_1, z_2, we have that, $z_1 + z_2 = 2 + \lambda$, and $z_1 z_2 = 1$.

2.3.1 Generalized Eigenvalue Form

To proceed further, let $\mathbf{u} = [u(0), u(1), \ldots, u(\tau)] \in \mathbb{R}^{\tau+1}$ be a non-zero vector, and let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{\tau+1\times\tau+1}$. Then (18) can be written as $\mathcal{B} = \mathcal{S} + \lambda(\mathcal{Q} - \tilde{c})$, where

$$\mathcal{S} = \sum_{t=1}^{\tau} \big(u(t) - u(t-1)\big)^2 = \mathbf{u}^\intercal \mathbf{A}\mathbf{u}, \qquad \text{and} \qquad \mathcal{Q} = \left[\sum_{t=1}^{\tau} \tfrac{1}{2}\big(u^2(t) + u^2(t-1)\big)\right] = \mathbf{u}^\intercal \mathbf{B}\mathbf{u}, \tag{23}$$

with $\mathbf{B} = \mathrm{diag}\big([\tfrac{1}{2}, 1, 1, \ldots, 1, \tfrac{1}{2}]\big)$, and tridiagonal matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix}. \tag{24}$$

We can observe that $\mathcal{S}$ involves only the sum of squared differences, so it is always non-negative for any real vector $\mathbf{u}$. $\mathcal{S}$ is zero if and only if every difference in the sum is zero, which implies $\mathbf{u} = u(0)\mathbf{1}$, where $u(0)$ is a real constant and $\mathbf{1} = [1, 1, \ldots, 1]$. However, constant $\mathbf{u}$ is not admissible due to the endpoint constraints. Therefore, $\mathbf{A}$ is a symmetric positive semi-definite matrix with a one-dimensional null-space spanned by the all-ones vector. Similarly, $\mathcal{Q}$ is a sum of squares with strictly positive weights. $\mathcal{Q}$ is zero only if $\mathbf{u} = 0$. Converting this sum to matrix form corresponds to $\mathbf{B}$ being diagonal with strictly positive entries, so $\mathbf{B}$ is a symmetric positive definite matrix.

Define $\tilde{\lambda} = -\lambda$. By taking the gradient of $\mathcal{B}$ with respect to $\mathbf{u}$ and setting to zero, we have $\mathbf{A}\mathbf{u} = \tilde{\lambda}\,\mathbf{B}\mathbf{u}$, which is equivalent to $\mathbf{B}^{-1}\mathbf{A}\mathbf{u} = \tilde{\lambda}\mathbf{u}$. This implies that

$$\tilde{\lambda} = \frac{\mathbf{u}^\intercal \mathbf{A}\mathbf{u}}{\mathbf{u}^\intercal \mathbf{B}\mathbf{u}}, \tag{25}$$

is the generalized eigenvalue of $\mathbf{A}$ with respect to $\mathbf{B}$, and $\mathbf{u} \neq 0$ is the corresponding generalized eigenvector of the pair $(\mathbf{A}, \mathbf{B})$. Because $\mathbf{A}$ is symmetric positive semi-definite and $\mathbf{B}$ is symmetric positive definite, the generalized eigenvalues $\tilde{\lambda}$ of the system $\mathbf{A}\mathbf{u} = \tilde{\lambda}\,\mathbf{B}\mathbf{u}$ are real and non-negative. A constant vector of ones lies in the null-space of $\mathbf{A}$ so that $\tilde{\lambda} = 0$ is the eigenvalue corresponding to no variation (all entries in $\mathbf{u}$ are equal). Such eigenvector violates the endpoint constraints $\delta(0) = 1$, $\delta(\tau) = 0$, so it is not admissible.

Therefore in (18), to minimize $\mathcal{S}$ subject to $\mathcal{Q} = \tilde{c}$, the optimal $\mathbf{u}$ must be a generalized eigenvector of $(\mathbf{A}, \mathbf{B})$ associated with the smallest positive eigenvalue of $\mathbf{B}^{-1}\mathbf{A}$.

We now show that for all $\mathbf{u} \neq 0$, $0 < \tilde{\lambda} < 4$. Using the elementary mean-squared inequality that $\forall x_1, x_2 \in \mathbb{R}$, $0 \leq (x_1 - x_2)^2 \leq 2(x_1^2 + x_2^2)$ holds with equality if and only if $x_1 = x_2 = 0$. Applying this for $x_1 = u(t)$, and $x_2 = u(t-1)$ then summing both sides over the window yields

$$0 \leq \sum_{t=1}^{\tau} \big(u(t) - u(t-1)\big)^2 \leq 4 \sum_{t=1}^{\tau} \tfrac{1}{2}\big(u^2(t) + u^2(t-1)\big) \tag{26}$$

Using the matrix form in (23), this is equivalent to $0 \leq \mathbf{u}^\intercal \mathbf{A}\mathbf{u} \leq 4\,\mathbf{u}^\intercal \mathbf{B}\mathbf{u}$. For $\mathbf{u} \neq 0$, strict inequality holds, so that $0 < \mathbf{u}^\intercal \mathbf{A}\mathbf{u} < 4\,\mathbf{u}^\intercal \mathbf{B}\mathbf{u}$, and therefore

$$0 < \frac{\mathbf{u}^\intercal \mathbf{A}\mathbf{u}}{\mathbf{u}^\intercal \mathbf{B}\mathbf{u}} < 4. \tag{27}$$

Comparing with (25) shows that for all $\mathbf{u} \neq 0$, $0 < \tilde{\lambda} < 4$. The optimal value of the Lagrange multiplier $\lambda = -\tilde{\lambda}$ must therefore be in the range $-4 < \lambda < 0$. This corresponds to the case when the two roots of the characteristic equation $z^2 - (2 + \lambda)\,z + 1 = 0$ are complex conjugates on the unit circle, $z_{1,2} = e^{\pm j\theta}$, for some $\theta \in (0, \pi)$. From the trace of the roots, we have $z_1 + z_2 = e^{j\theta} + e^{-j\theta} = 2 + \lambda$, and solving for λ yields

$$\lambda = -2(1 - \cos(\theta)) = -4\sin^2\left(\frac{\theta}{2}\right) \tag{28}$$

2.3.2 Complete solution

The general solution to (22) is of the form $u(t) = \bar{p}_1 z_1^t + \bar{p}_2 z_2^t$, which after substituting the roots, and absorbing $\bar{p}_1, \bar{p}_2$ into real constants p_1, p_2 leads to

$$u(t) = p_1 \cos(t\,\theta) + p_2 \sin(t\,\theta). \tag{29}$$

From (29), we have

$$\begin{aligned} u(0) &= p_1, \\ u(1) &= p_1 \cos(\theta) + p_2 \sin(\theta). \end{aligned} \tag{30}$$

Plugging (30) and (28) into the initial boundary condition identity (21) leads to $p_1 \cos(\theta) + p_2 \sin(\theta) - p_1 = p_1 \cos(\theta) - p_1$. This is satisfied if and only if $p_2 \sin(\theta) = 0$. For any non-trivial $\theta \neq 0$, and $\sin(\theta) \neq 0$, this implies $p_2 = 0$. Therefore, for all θ, we must have $p_2 = 0$, which simplifies (29) to

$$u(t) = p_1 \cos(t\,\theta). \tag{31}$$

Next, plugging (31) into the final boundary condition (20), we obtain the identity $\sin(\theta) \sin((\tau)\theta) = 0$. For any non-trivial $\theta \in (0, \pi)$, such that $\sin(\theta) \neq 0$, this implies that $(\tau)\theta = m\pi$ for any integer $m = 0, 1, 2, \ldots, \tau$. Each m corresponds to the $\tau+1$ ordered eigenvalues $0 \leq \tilde{\lambda}_0 \leq \tilde{\lambda}_1 \leq \ldots \leq \tilde{\lambda}_\tau < \infty$ representing the frequency modes of $\mathbf{B}^{-1}\mathbf{A}$. The integer $m = 1$ would correspond to $\tilde{\lambda}_1$, the smallest positive eigenvalue of $\mathbf{B}^{-1}\mathbf{A}$ which minimizes the regularized objective. Therefore, we have

$$\theta = \frac{\pi}{\tau}. \tag{32}$$

Plugging (32) into (31) and (28) leads to the complete solution

$$\begin{aligned} u(t) &= u(0) \cos\bigl(\pi\, t/\tau\bigr), \\ \lambda &= -4 \sin^2\bigl(0.5\,\pi/\tau\bigr) \approx -\pi^2 \tau^{-2}, \end{aligned} \tag{33}$$

and in general, for $m = 0, 1, 2 \ldots, \tau$, each eigenvalue of $\mathbf{B}^{-1}\mathbf{A}$ is of the form $\tilde{\lambda}_m = 4 \sin^2\bigl(0.5\, m\pi/\tau\bigr)$. Evaluating the energy constraint term $\sum_{t=1}^{\tau} \frac{1}{2}\bigl(u^2(t) + u^2(t-1)\bigr) = \tilde{c}$ leads to $u^2(0)\frac{\tau}{2} = \tilde{c}$, and therefore $u(0) = \sqrt{\frac{2\tilde{c}}{\tau}}$. For the optimal $\mathbf{u}$, $\mathcal{Q}^* = \tilde{c}$, so $\mathcal{B}^* = \mathcal{S}^* = \sum_{t=1}^{\tau}\bigl(u(t) - u(t-1)\bigr)^2$ yields

$$\mathcal{B}^* = 4\, u^2(0) \tfrac{\tau}{2} \sin^2\bigl(\tfrac{\pi}{2\tau}\bigr) = -\tilde{c}\,\lambda. \tag{34}$$

Finally, plugging in the corresponding boundary conditions for $\delta(t)$ given by $\delta(0) = \mu$, $\delta(\tau) = 0$ by using (33) in $\delta(t) = a + b\,u(t)$ (16) we get

$$\begin{aligned} \delta(0) &= a + b\,u(0) \\ \delta(\tau) &= a + b\,u(\tau) = a - b\,u(0) \end{aligned} \tag{35}$$

Solving the simultaneous equation in (35) for a, b leads to the constants $a = \frac{\mu}{2}$, and $b = \frac{a}{u(0)}$. Substituting a and b in (16), finally leads to

$$\delta(t) = \mu\, \frac{1}{2}\Bigl[1 + \cos\bigl(\tfrac{\pi\, t}{\tau}\bigr)\Bigr] = \mu \cos^2\bigl(\tfrac{\pi\, t}{2\tau}\bigr). \tag{36}$$

Therefore, the regularized minimization problem (19) leads to (36) a half-period second-order raised cosine function commonly known as the *cosine annealing* schedule. To check, the trust-region constraint is always satisfied since from (36), $\delta^2(t) \leq \mu^2$, which implies $\rho = \mu$.

Similarly, applying the same regularized Dirichlet energy minimization to $\delta^2(t)$ in (13) yields

$$\delta(t) = \mu \cos\bigl(\tfrac{\pi\, t}{2\tau}\bigr). \tag{37}$$

2.4 Structural Properties of the Dirichlet Energy Minimizing Window Functions

The unique solutions that arise from the three variational problems posed in (8), (13), and (18) share several properties. Their boundary conditions of the trust-region are well-defined, the trust-region vanishes towards the end of the learning window. Each solution can also be written in a common structural form

$$\delta(t) = \mu\, \psi(t), \tag{38}$$

where $0 \leq \psi(t) \leq 1$, $\psi(0) = 1$, $\psi(\tau) = 0$, and $\psi(t) \to 0$ as $t \to \tau$. The function $\psi(t)$ decays monotonically over the interior of the finite window $t \in \{0, \dots, \tau\}$ that governs the variation of $\delta^2(t) = \mathbb{E}[\Delta^2(t+1, i)]$.

Intuitively, forcing $\delta(t) \to 0$ at the end of the window mirrors classical trust-region methods, where the trust-region radius is tightened as the parameter iterates approach a candidate solution. In the stochastic setting, this encourages smaller parameter updates near the end of the window, so that if $\mathrm{w}(t+1, i)$ is already in the neighborhood of a local optimum, the iterative trust-region is tightened around that neighborhood.

Furthermore, the average of the variable trust-region radius and its squared radius over the learning window are respectively

$$\frac{1}{\tau}\sum_{t=0}^{\tau-1} \delta(t) = \frac{\mu}{\tau}\sum_{t=0}^{\tau-1} \psi(t), \qquad \text{and} \qquad \frac{1}{\tau}\sum_{t=0}^{\tau-1} \delta^2(t) = \frac{\mu^2}{\tau}\sum_{t=0}^{\tau-1} \psi^2(t). \tag{39}$$

For the window functions derived, (12), (15), (36) and (37), as $\tau \to \infty$, the average $\frac{\mu}{\tau}\sum_{t=0}^{\tau-1} \psi(t)$ converges to a strictly positive constant (specifically, $\frac{\mu}{2}$ for the linear window and second-order raised cosine window, $\frac{2}{3}\mu$ for the square-root linear window, and $\frac{2}{\pi}\mu$ for the first-order raised cosine window). Also, since $0 \leq \psi^2(t) \leq \psi(t) \leq 1$, we have $0 < \frac{1}{\tau}\sum_{t=0}^{\tau-1} \psi^2(t) < \frac{1}{\tau}\sum_{t=0}^{\tau-1} \psi(t)$. Consequently, despite $\delta^2(\tau) = 0$, the average of the squared trust-region radius $\frac{1}{\tau}\sum_{t=0}^{\tau-1} \delta^2(t)$ remains bounded away from zero. In this sense, the window functions avoid an overly aggressive decay of the learning-rate while still enforcing a vanishing trust-region at the end of the window.

In summary, the variational problems and their solutions correspond to three distinct notions of smoothness for the vanishing trust-region radius. Linear decay (12) arises as the unique solution to the Dirichlet energy minimization problem defined on the trust-region radius $\delta(t)$ itself (8), and therefore corresponds to the function $\delta(t)$ with minimal total variation over the learning window. In contrast, square-root linear decay (15) is the solution to the Dirichlet energy minimization problem defined on the squared radius $\delta^2(t)$ (13), and corresponds to the function $\delta(t)$ whose squared radius exhibits minimal total variation over the learning window. As a consequence, this solution maintains larger values over a greater portion of the learning window, followed by a more rapid contraction near the end of the window. Conversely, the raised-cosine (cosine annealing) functions (36) and (37) arise as solutions to regularized Dirichlet energy minimization problems posed respectively on $\delta(t)$ (18) and $\delta^2(t)$. The quadratic regularization induces smoother transitions, characterized by a slower initial decay and a sharper contraction as $t \to \tau$. These window functions, therefore correspond to extremal solutions of distinct first-order variational problems, each encoding a different notion of smoothness in the evolution of the variable trust-region radius.

2.5 On the role of Boundary values

The appearance of boundary values in the present framework follows as a direct consequence of the vanishing trust-region formulation. The trust-region radius is required to begin at some initial value and evolve to a terminal value over a finite learning window. Consequently, the unknown function is not free to vary arbitrarily, but must evolve between fixed endpoints. The variational energy functional then selects, among all trust-region trajectories satisfying these endpoint conditions, the one with minimum energy. This is analogous to the structure encountered in classical variational theory, where the solution is determined jointly by an energy functional and its associated boundary conditions (Iwaniec & Onninen, 2022; Grady & Polimeni, 2010).

In this sense, learning-rate schedules arise as solutions to trust-region variational boundary-value problems. The energy functional determines the smoothness properties of the trajectory, while the boundary values determine the class of admissible trust-region evolutions. Different schedules therefore correspond to different solutions of the same underlying variational formulation and boundary conditions.

Next, we derive general forms of these energy minimizing solutions subject to fixed-point boundary values.

3 Generalized forms

Let $1 \leq p < \infty$, and define the first-order variational energy minimization functional

$$\min_{0 \leq \delta^p(t) \leq \mu^p} \sum_{t=1}^{\tau} \left(\delta^p(t) - \delta^p(t-1)\right)^2, \tag{40}$$

Differentiating with respect to $\delta^p(i)$ for any interior index $i \in \{1, \ldots, \tau - 1\}$, and setting to zero results in the second-order homogeneous linear difference equation with constant coefficients

$$\delta^p(i+1) - 2\,\delta^p(i) + \delta^p(i-1) = 0. \tag{41}$$

The corresponding characteristic equation is $z^2 - 2\,z + 1 = (z-1)^2 = 0$ which has 1 as a root with multiplicity 2. Therefore, the general solution to (41) must be a polynomial in t, of at most degree 1 (Elaydi, 2005; Lueker, 1980),

$$\delta^p(t) = a_0 + a_1\,t. \tag{42}$$

A complete solution can then be obtained by plugging in at least $q \geq 2$ fixed boundary values: the fixed terminal value, and at least one fixed initial value, leading to $q - 1$ sub-interval(s) of the learning window.

3.1 2-point boundary value

Consider the initial and terminal boundary values: $\delta^p(0) = \mu^p$, $\delta^p(\tau) = 0$ over the single sub-interval $0 \leq t \leq \tau$ Solving (42), subject to $\delta^p(0) = \mu^p$, $\delta^p(\tau) = 0$ gives $a_0 = \mu^p$, and $a_1 = -\mu^p/\tau$. Therefore, the complete solution to (41) subject to the 2-point boundary value is

$$\delta^p(t) = \mu^p \left(1 - \frac{t}{\tau}\right), \quad 0 \leq t \leq \tau, \tag{43}$$

which is known as the linear decay schedule. We can simplify this to a normalized functional form. Let $r = t/\tau$, define an input function $x(r) \coloneqq r \in [0, 1]$, then

$$\delta^p(t) = \mu^p \left(1 - x(r)\right), \quad 0 \leq r \leq 1. \tag{44}$$

3.2 3-point boundary value

Consider the three fixed boundary values: $\delta^p(0) = 0$, $\delta^p(k) = \mu^p$, $\delta^p(\tau) = 0$, where $0 \leq k < \tau$. This corresponds to two sub-intervals. On the first interval over $0 \leq t \leq k$, the solution is obtained by solving (42) subject to $\delta^p(0) = 0$, $\delta^p(k) = \mu^p$. On the second interval over $k \leq t \leq \tau$, the solution is obtained by solving (42) subject to $\delta^p(k) = \mu^p$, $\delta^p(\tau) = 0$. Combining the piecewise solutions, the complete solution to (41) subject to the 3-point boundary value is

$$\delta^p(t) = \begin{cases} \mu^p\,\frac{t}{k}, & 0 \leq t \leq k, \quad k > 0 \\ \mu^p\,\frac{\tau - t}{\tau - k} & k \leq t \leq \tau, \quad k \geq 0. \end{cases} \tag{45}$$

This can be simplified further. Let $r = t/\tau \in [0, 1]$, a normalized rise-time $m = k/\tau \in [0, 1)$, then the reparameterized input function $x(r; m) \in [0, 1]$ for (45) is

$$x(r; m) \coloneqq \begin{cases} r, & m = 0, \\ \max\{\frac{m - r}{m},\ \frac{r - m}{1 - m}\}, & 0 < m < 1, \end{cases} \tag{46}$$

and equivalently, (45) becomes

$$\delta^p(t) = \mu^p \left(1 - x(r;m)\right), \quad 0 \le r \le 1, \quad 0 \le m < 1. \tag{47}$$

For its first sub-interval, the hat-shaped (or triangular-like) mapping (45) corresponds to a warmup schedule that reaches μ^p at time m, and then a decaying schedule to zero in the second sub-interval. Importantly, both (44) and (47) share the same underlying functional form. When $m = 0$, it follows that $x(r;m) = x(r;0) = r$, hence (44) is a special case of (47) when $m = 0$.

3.3 4-point boundary value

Consider the four fixed boundary values: $\delta^p(0) = 0$, $\delta^p(k) = \mu^p$, $\delta^p(k') = \mu^p$, $\delta^p(\tau) = 0$, where $0 \le k \le k' < \tau$. This corresponds to three sub-intervals. On the first interval over $0 \le t \le k$, the solution is obtained by solving (42) subject to $\delta^p(0) = 0$, $\delta^p(k) = \mu^p$. On the second interval over $k \le t \le k'$, the solution is obtained by solving (42) subject to $\delta^p(k) = \mu^p$, $\delta^p(k') = \mu^p$. Finally, for the third interval over $k' \le t \le \tau$, the solution is obtained by solving (42) subject to $\delta^p(k') = \mu^p$, $\delta^p(\tau) = 0$. Combining the three piecewise solutions, the complete solution to (41) subject to the 4-point boundary value, results in

$$\delta^p(t) = \begin{cases} \mu^p \, \frac{t}{k}, & 0 \le t \le k, \quad k > 0 \\ \mu^p & k \le t \le k', \quad k \ge 0 \\ \mu^p \, \frac{\tau - t}{\tau - k'} & k' \le t \le \tau, \quad k' \ge 0. \end{cases} \tag{48}$$

Let $r = t/\tau \in [0,1]$, and let $m = k/\tau \in [0,1)$ indicate the rise time and subsequently, the start of a plateau of width $\varepsilon = (k' - k)/\tau \in [0,1)$, so that $k = m\tau$, and $k' = (m + \varepsilon)\tau$. Then the reparameterized input function $x(r;m,\varepsilon) \in [0,1]$ for (48) is

$$x(r;m) := \begin{cases} \max\{0, \frac{r-\varepsilon}{1-\varepsilon}\}, & m = 0, \\ \max\{\frac{m-r}{m}, 0, \frac{r-(m+\varepsilon)}{1-(m+\varepsilon)}\}, & 0 < m < 1, \end{cases} \tag{49}$$

and equivalently (48) can be expressed as

$$\delta^p(t) = \mu^p \left(1 - x(r;m,\varepsilon)\right), \quad 0 \le r \le 1, \quad 0 \le m < 1. \tag{50}$$

The resulting function (48) then corresponds in the first sub-interval to a warmup schedule that rises to μ^p, and then a plateau in the second sub-interval, followed by a decaying schedule to zero in the third sub-interval. This leads a trapezoidal shape, and exactly a warmup-stable-decay schedule.

Again, observe that for the case $m = 0, \varepsilon = 0$, it follows that $x(r;m,\varepsilon) = x(r;0,0) = u$, therefore (44) is a special case of (50). Similarly, when $m \ne 0, \varepsilon = 0$, it follows that $x(r;m,\varepsilon) = x(r;m,0) = x(r;m)$, therefore (47) is a special case of (50). Hence, both (44), (47) and (50) share the same underlying functional form, with (50) having a more general input reparameterization function.

Finally, denote $\delta^p(t) = \mu^p \, \phi(t)$, where $\phi(t) = 1 - x(r;m,\varepsilon)$, then

$$\delta(t) = \mu \, \psi(t) = \mu \, \phi^{\frac{1}{p}}(t) \tag{51}$$

is the unique minimizer of the first-order total variational energy functional, which satisfies $\delta(t) \le \mu$, $\delta(t) \to 0$ as $t \to \tau$ subject to $2 \le q \le 4$ fixed boundary points.

Likewise, instead of (40), applying the regularized first-order total variational energy functional to each subinterval of the multi-point boundary-value problem yields the raised-cosine family

$$\phi(t) = \cos^2\left(0.5 \, \pi \, x(r;m,\varepsilon)\right), \tag{52}$$

for $2 \le q \le 4$ fixed boundary points.

Remarks: The preceding analysis shows that many commonly used learning-rate schedules arise naturally as solutions to a trust-region variational problem. By treating $\delta^p(t)$ as a varying trust-region radius and minimizing its total variation subject to fixed boundary values, we obtain schedules that evolve as smoothly as possible, while still satisfying the asymptotic requirement $\delta(t) \to 0$ as $t \to \tau$.

The resulting p-th root linear decay, warmup-decay, and warmup-stable-decay schedules are illustrated in Figure 1. They correspond respectively to 2-point, 3-point, and 4-point boundary-value problems, yet all share a common variational structure. Consequently, learning-rate schedules may be interpreted as vanishing trust-region functions that progressively tighten the admissible trust-region of the update step over the learning horizon. Under this interpretation, schedule design is no longer an independent heuristic, but rather a principled consequence of solving a first-order variational optimization problem subject to trust-region boundary constraints. More generally, alternative variational functionals would produce different classes of schedules, each reflecting the structure of the underlying variational principle.

Beyond learning-rate scheduling , such as recovering the warmup-stable-decay, and constant-cooldown schedules produced in Hägele et al. (2024), the resulting window functions are also closely related to classical tapering functions used in spectral analysis (Prabhu, 2018; Nuttall, 1981; Blackman & Tukey, 1958). In that setting, the shape of a window determines its spectral smoothing properties. This follows from the fact that multiplying an iteration-dependent quantity by $\delta(t)$ in the time domain corresponds, in the Fourier domain, to convolving its spectrum with the Fourier transform of $\delta(t)$ (Harris, 1978). Under appropriate boundary conditions, the present framework can similarly recover classical taper shapes, including the Hann window (Nuttall, 1981) and the Tukey window (Harris, 1978).

3.4 Smoothness Factor

Recall that $\delta^p(t) = \mu^p\,\phi(t)$, for a finite $p \geq 1$. Consider a continuously differentiable (smooth) prototype function $\varphi : [0,1] \to [0,1]$ defined over a fixed window of iterations $t \in \{0,\dots,\tau\}$ by

$$\phi(t) = \varphi\big(\tfrac{t}{\tau}\big)\,. \tag{53}$$

For the interior $t \in \{1,\dots,\tau-1\}$, let $x(t) = \frac{t}{\tau}$ denote an ordered uniform discretization of $(0,1)$ over which $\varphi(x)$ decays monotonically. Assume further that $\varphi'(x)$, the first-derivative of φ with respect to x exists for all $x \in (0,1)$ bounded as the fixed window length τ grows. The first-order total variational energy functional (40) yields

$$\mathcal{D}_{\tau,\phi(t)} = \mu^{2p} \sum_{t=1}^{\tau} \big(\phi(t) - \phi(t-1)\big)^2. \tag{54}$$

Further, for any $t \in \{1,\dots,\tau-1\}$, by the mean-value theorem, there exists a $\bar{x} \in \big[\frac{t-1}{\tau}, \frac{t}{\tau}\big]$, such that $\phi(t) - \phi(t-1) = \varphi\big(\frac{t}{\tau}\big) - \varphi\big(\frac{t-1}{\tau}\big) = \varphi'(\bar{x})\,\big(\frac{t}{\tau} - \frac{t-1}{\tau}\big) = \frac{1}{\tau}\,\varphi'(\bar{x})$. Plugging this into (54), we obtain

$$\mathcal{D}_{\tau,\phi(t)} = \left(\frac{\mu^p}{\tau}\right)^2 \sum_{t=1}^{\tau} \big(\varphi'(\bar{x})\big)^2, \tag{55}$$

Define,

$$L = \frac{1}{\tau} \sum_{t=1}^{\tau} \big(\varphi'(\bar{x})\big)^2. \tag{56}$$

Then (55) can be re-expressed in terms of (56) as (55) becomes

$$\mathcal{D}_{\tau,\phi(t)} = \mu^{2p}\,\tau^{-1}\,L, \tag{57}$$

where the factor L separates the contribution of $\phi(t)$ to the Dirichlet energy from the multiplicative scaling term $\mu^{2p}\,\tau^{-1}$. Moreover, since L is the average value of $|\varphi'(\bar{x}))|^2$, therefore smaller values of L correspond to smoother variations, whereas larger values indicate more rapid variations over the window. Furthermore,

$$\lim_{\tau\to\infty} L = \int_0^1 |\varphi'(x)|^2\,dx, \tag{58}$$

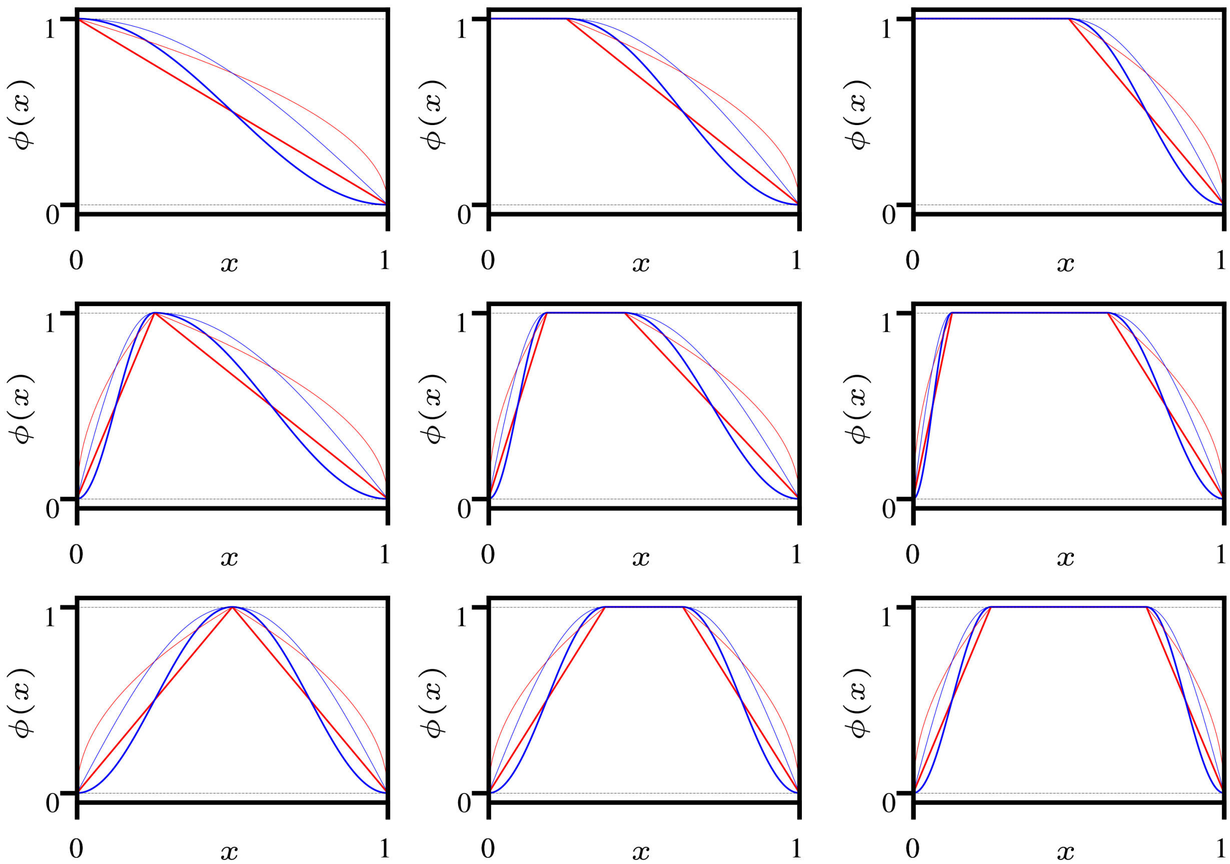

Figure 1: Window shapes obtained by passing $x(r;m,\varepsilon)$ into the linear function $1-x$, and raised-cosine function $\cos^2\big(0.5\,\pi\,x\big)$ for $p=1$ and $p=2$. Linear decay $x\big(r;0,0\big)$, $p=1$ (thick, red), square-root linear decay $x\big(r;0,0\big)$, $p=2$ (thin, red color), second-order raised-cosine $x\big(r;0,0\big)$, $p=1$ (thick, blue), and first-order raised-cosine $x\big(r;0,0\big)$, $p=2$ (thin, blue). **Top row**: $m=0$, $\varepsilon=\{0,0.25,0.5\}$ (Left to Right). **Second row**: $m=0.25$, $\varepsilon=\{0,0.25,0.5\}$ (Left to Right). **Third row**: $m=0.5$, $\varepsilon=\{0,0.25,0.5\}$ (Left to Right). Default decay shape (first row, first column); Half-Trapezoidal shapes (red, first row, second–third columns); Triangular shapes (red, second–fourth rows, first column); Full-Trapezoidal shapes (red, second–fourth rows, second–third columns); Hann-window shape (blue, third row, first column); Tuckey-window shapes (blue, third row, second–third columns).

showing that L converges to the continuous Dirichlet energy of φ. In the limit $\tau \to \infty$, L becomes independent of the particular choice of scaling parameters μ, p, and τ. Consequently, the factor L may be viewed as the normalized Dirichlet-energy measure of the smoothness of $\phi(t)$ over the unit interval $[0, 1]$.

Moreover, since $\mathcal{D}_{\tau,\phi(t)} = \mu^{2p}\tau^{-1}L$, the minimization of the Dirichlet energy is equivalent to the minimization of the smoothness factor L. Consequently, among all admissible window functions satisfying the same boundary-values, the variational solutions derived earlier attain the smallest possible value of L and therefore represent the smoothest trust-region radius trajectories under their corresponding variational formulations.

3.5 Vanishing Window Ratios

We now make precise the relationship between the ratio $\delta(t)/\delta(t-1) = \psi(t)/\psi(t-1)$ and the stability of the parameter update step $\Delta(t+1, i)$.

First, we will show that if the ratio $\delta(t)/\delta(t-1) \to 0$ then we must have $\delta(t) \to 0$. Next, we show that $\delta(t) \to 0$ strictly faster than $\delta(t)/\delta(t-1) \to 0$. Finally, we show that these vanishing window ratios induce a stronger exponential contraction of the update step.

For this proof, assume that $0 \le \delta(t) \le 1$ and that $\delta(t)/\delta(t-1) \to 0$ as $t \to \tau$. Therefore, by definition of a limit, there exists a fixed $\epsilon \in (0, 1)$ and an index t' with $0 < t' \le \tau$ in the window, such that $\delta(t)/\delta(t-1) \le \epsilon, \forall t \ge t'$. This inequality implies $\delta(t'+1) \le \epsilon\, \delta(t')$. Iterating forward, for any integer $k \ge 1$ with $t' + k \le \tau$ we obtain

$$\delta(t'+k) \le \epsilon^k\, \delta(t'). \tag{59}$$

Because $\epsilon \in (0, 1)$, we must have $\epsilon^k \to 0$ as $k \to \infty$, and so in (59) $\delta(t'+k) \to 0$ as $t' + k \to \tau$. Therefore, $\delta(t)/\delta(t-1) \to 0$ as $t \to \tau$ implies $\delta(t) \to 0$ as $t \to \tau$.

In addition, because $0 \le \delta(t) \le 1$, the monotonically decreasing sequence $0 \le \delta(t') \le \delta(t'-1) \le 1$ directly implies $0 \le \delta(t')/\delta(t'-1) \le 1$, and that $\delta(t')\,\delta(t'-1) < \delta(t')$ from which we have

$$\delta(t') \le \frac{\delta(t')}{\delta(t'-1)} \le \epsilon. \tag{60}$$

The inequality in (60) means that the ratio $\delta(t')/\delta(t'-1)$ is always larger than $\delta(t')$, with equality only at $t' = \tau$ where $\delta(\tau) = 0$. Combining with $\delta(t)/\delta(t-1) \le \epsilon, \forall\, t \ge t'$ and (59), then for any $t = t' + k$, we have that $\delta(t)$ decays at a rate proportional to $\epsilon^{t-t'}$ while the ratio $\delta(t)/\delta(t-1)$ only decays linearly to zero at a rate proportional to ϵ. Therefore, we have that $\delta(t) \to 0$ strictly faster than $\delta(t)/\delta(t-1) \to 0$.

Finally, from the trust-region definition $\mathbb{E}[\Delta^2(t+1, i)] = \delta^2(t)$, it follows immediately that $\delta(t) \to 0$ implies $\mathbb{E}[\Delta^2(t+1, i)] \to 0$. Furthermore, from (59),

$$\delta(t) \le \epsilon^{\,t-t'}\delta(t'), \tag{61}$$

for all $t \ge t'$, and hence the expected squared parameter update sequence satisfies

$$\mathbb{E}[\Delta^2(t+1, i)] = \delta^2(t) \le \epsilon^{2\,(t-t')}\delta^2(t'). \tag{62}$$

and therefore contracts exponentially towards zero. Since $0 < \epsilon < 1$, smaller values of the vanishing window ratio $\delta(t)/\delta(t-1)$ induce a stronger exponential contraction of the update step. Consequently, vanishing window ratios improve the asymptotic stability of the parameter update dynamics.

In the special case $t' = 0$, where $\delta(0) = \mu$, we obtain

$$\mathbb{E}[\Delta^2(t+1, i)] \le \mu^2\epsilon^{2t}, \tag{63}$$

which provides an explicit exponential upper bound in terms of the initial trust-region radius μ and contraction factor $\epsilon \in (0, 1)$. Consequently,

$$\mathbb{E}\big[\|\Delta(t+1)\|_2^2\big] = \sum_{i=1}^{n} \mathbb{E}[\Delta^2(t+1, i)] \le n\,\mu^2\epsilon^{2t}, \tag{64}$$

showing that the parameter update step contracts exponentially to zero in the mean-square sense. Therefore, vanishing trust-region window induces exponential mean-square stability of the update dynamics.

4 Conclusion

This paper developed a variational trust-region framework for stochastic gradient learning. By formulating the update step as a trust-region problem, we obtained an iteration-dependent learning rate whose fixed-radius form recovers the core structure of adaptive moment estimation. Extending the framework to a varying trust-region radius led naturally to variational problems posed on the evolution of the trust-region itself.

Within this setting, linear decay, cosine annealing, and their corresponding p-th root forms arise as unique solutions of Dirichlet and regularized Dirichlet energy minimization problems. More generally, the derivations reveal that learning-rate schedules can be interpreted as solutions of trust-region variational boundary-value problems. The resulting framework extends naturally to multi-point boundary-value formulations, yielding warmup-decay, warmup-stable-decay, and related hybrid schedules within the same mathematical structure.

More broadly, learning-rate annealing can be interpreted within the class of total variational boundary-value problems. This places learning-rate scheduling within a well-established mathematical framework in which solutions are determined jointly by an energy functional and fixed boundary values. Hence, this work provides a unified formulation connecting adaptive moment estimation, learning-rate annealing, and trust-region evolution in stochastic gradient methods.

References

Naman Agarwal, Surbhi Goel, and Cyril Zhang. Acceleration via Fractal Learning Rate Schedules. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 87–99, Virtual, USA, July 2021. PMLR. URL `https://proceedings.mlr.press/v139/agarwal21a.html`.

Irwan Bello, William Fedus, Xianzhi Du, Ekin Dogus Cubuk, Aravind Srinivas, Tsung-Yi Lin, Jonathon Shlens, and Barret Zoph. Revisiting ResNets: Improved training and scaling strategies. In *35th Annual Conference on Neural Information Processing Systems*, Virtual, USA, December 2021. Curran Associates, Inc. URL `https://openreview.net/forum?id=dsmxf7FKiaY`.

Shane Bergsma, Nolan Simran Dey, Gurpreet Gosal, Gavia Gray, Daria Soboleva, and Joel Hestness. Straight to zero: Why linearly decaying the learning rate to zero works best for LLMs. In *The Thirteenth International Conference on Learning Representations*, Singapore, October 2024. URL `https://openreview.net/forum?id=hrOlBgHsMI`.

R. B. Blackman and J. W. Tukey. The measurement of power spectra from the point of view of communications engineering — Part I. *The Bell System Technical Journal*, 37(1):185–282, January 1958. ISSN 0005-8580. doi: 10.1002/j.1538-7305.1958.tb03874.x.

Léon Bottou, Frank E. Curtis, and Jorge Nocedal. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, May 2018.

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 9650–9660, NashviIle, TN, 2021. URL `https://openaccess.thecvf.com/content/ICCV2021/html/Caron_Emerging_Properties_in_Self-Supervised_Vision_Transformers_ICCV_2021_paper.html`.

Saber Elaydi. *An Introduction to Difference Equations*. Springer, New York, 2005. ISBN 978-0-387-27602-1.

Rong Ge, Sham M. Kakade, Rahul Kidambi, and Praneeth Netrapalli. Rethinking learning rate schedules for stochastic optimization. In *International Conference on Learning Representations*, New Orleans, LA, September 2018. URL `https://openreview.net/forum?id=HJePy3RcF7`.

Baptiste Goujaud, Damien Scieur, Aymeric Dieuleveut, Adrien B. Taylor, and Fabian Pedregosa. Super-acceleration with cyclical step-sizes. In *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*, pp. 3028–3065, Virtual, 2022. PMLR. URL `https://proceedings.mlr.press/v151/goujaud22a.html`.

Leo J. Grady and Jonathan R. Polimeni. *Discrete Calculus.* Springer, London, 2010. ISBN 978-1-84996-289-6 978-1-84996-290-2. doi: 10.1007/978-1-84996-290-2.

Alex Graves, Abdel-rahman Mohamed, and Geoffrey Hinton. Speech recognition with deep recurrent neural networks. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 6645–6649, Vancouver, BC, Canada, 2013. IEEE. doi: 10.1109/ICASSP.2013.6638947.

Alexander Hägele, Elie Bakouch, Atli Kosson, Loubna Ben Allal, Leandro Von Werra, and Martin Jaggi. Scaling Laws and Compute-Optimal Training Beyond Fixed Training Durations. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, Vancouver, Canada, November 2024. URL `https://openreview.net/forum?id=Y13gSfTjGr&noteId=jbPUoMWzI4`.

F.J. Harris. On the use of windows for harmonic analysis with the discrete Fourier transform. *Proceedings of the IEEE*, 66(1):51–83, January 1978. ISSN 1558-2256. doi: 10.1109/PROC.1978.10837.

Tong He, Zhi Zhang, Hang Zhang, Zhongyue Zhang, Junyuan Xie, and Mu Li. Bag of tricks for image classification with convolutional neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 558–567, Long Beach, CA, 2019. URL `https://openaccess.thecvf.com/content_CVPR_2019/html/He_Bag_of_Tricks_for_Image_Classification_with_Convolutional_Neural_Networks_CVPR_2019_paper.html`.

Gao Huang, Yixuan Li, Geoff Pleiss, Zhuang Liu, John E. Hopcroft, and Kilian Q. Weinberger. Snapshot ensembles: Train 1, get M for free. In *5th International Conference on Learning Representations*, Toulon, France, April 2017. URL `https://openreview.net/forum?id=BJYwwY9ll`.

Tadeusz Iwaniec and Jani Onninen. The Dirichlet principle for inner variations. *Mathematische Annalen*, 383(1-2):315–351, June 2022. ISSN 0025-5831, 1432-1807. doi: 10.1007/s00208-020-02133-y.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *3rd International Conference on Learning Representations*, San Diego, CA, USA,, May 2015.

Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *5th International Conference on Learning Representations*, Toulon, France, April 2017. URL `https://openreview.net/forum?id=Skq89Scxx`.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *7th International Conference on Learning Representations*, New Orleans, LA, USA, May 2019. URL `https://openreview.net/forum?id=Bkg6RiCqY7`.

George S. Lueker. Some Techniques for Solving Recurrences. *ACM Comput. Surv.*, 12(4):419–436, December 1980. ISSN 0360-0300. doi: 10.1145/356827.356832.

Jorge Nocedal and Stephen J. Wright. Trust-Region Methods. In *Numerical Optimization*, pp. 66–100. Springer, New York, NY, 2006. ISBN 978-0-387-40065-5. doi: 10.1007/978-0-387-40065-5_4.

A. Nuttall. Some windows with very good sidelobe behavior. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 29(1):84–91, February 1981. ISSN 0096-3518. doi: 10.1109/TASSP.1981.1163506.

Neal Parikh and Stephen Boyd. Proximal Algorithms. *Foundations and Trends in Optimization*, 1(3): 127–239, January 2014. ISSN 2167-3888. doi: 10.1561/2400000003.

K. M. M. Prabhu. *Window Functions and Their Applications in Signal Processing.* CRC Press, Boca Raton, September 2018. ISBN 978-1-315-21638-6. doi: 10.1201/9781315216386.

Tijmen Tieleman and Geoffrey Hinton. Lecture 6.5-rmsprop: Divide the gradient by a running average of its recent magnitude. Technical report, COURSERA: Neural Networks for Machine Learning, 2012.

Hikaru Umeda and Hideaki Iiduka. Increasing Both Batch Size and Learning Rate Accelerates Stochastic Gradient Descent. *Transactions on Machine Learning Research*, November 2024. ISSN 2835-8856. URL `https://openreview.net/forum?id=sbmp55k6iE`.

Linwei Yue, Huanfeng Shen, Jie Li, Qiangqiang Yuan, Hongyan Zhang, and Liangpei Zhang. Image super-resolution: The techniques, applications, and future. *Signal Processing*, 128:389–408, November 2016. ISSN 0165-1684. doi: 10.1016/j.sigpro.2016.05.002.